

DW04_shad_reg

Shadow and Multibit Register

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

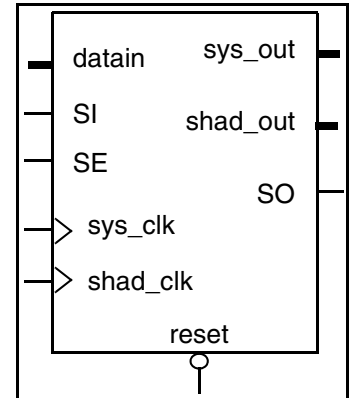
Features and Benefits

- Captures the state of system registers dynamically during system operation
- Serial access on shadow register to scan out the state of captured data
- Constructed with multi-bit flip-flop cells where possible; can be used as a simple, non-shadowed multi-bit register
- Parameterized width and number of registers (one or two)

Applications

- Diagnostics support for in-system testing
- Registers embedded for real-time emulation
- Computer, data, and telecommunications circuits that may benefit from a more efficient register implementation

Revision History



Description

DW04_shad_reg is a parameterized register pair. The first register is a parallel load, parallel output system register clocked on the `sys_clk` pin with an asynchronous reset controlled by the `reset` pin. The register is implemented in multi-bit (ganged) flip-flop cells if available in the target technology. The `datain` signal drives the DW04_shad_reg inputs and the output is `sys_out`.

The second register is a shadow register, which, like the system register, is also *width* bits wide and implemented in the same target technology cells. However, the shadow register is a parallel load shift register, which captures the output of the system register when sampled by `shad_clk`. This register outputs its contents to `shad_out` (parallel) and `SO` (serial).

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
datain	<i>width</i> bits	Input	Data input driving the input to the system register
sys_clk	1 bit	Input	Clock that samples the system register, positive edge triggered
shad_clk	1 bit	Input	Signal that clocks the output of the system register into the shadow register, positive edge triggered
reset	1 bit	Input	Asynchronous reset signal that clears the system and shadow registers
SI	1 bit	Input	Serial scan input, clocked by <code>shad_clk</code> when <code>SE</code> is high

Table 1-1 Pin Description (Continued)

Pin Name	Width	Direction	Function
SE	1 bit	Input	Serial scan enable, active high; enables scan only on the shadow register
sys_out	<i>width</i> bits	Output	Output of the system register
shad_out	<i>width</i> bits	Output	Parallel output of shadow register that lags system register by one cycle
SO	1 bit	Output	Serial scan output from shadow register When SE is low, represents the state of the MSB of the shadow register. When SE is high, each successive bit is shifted up one and SI is clocked into the LSB.

Table 1-2 Parameter Description

Parameter	Values	Description
width ¹	1 to 512 ^a	Defines width of system and shadow registers, and the input and output buses
bld_shad_reg	0 or 1	Defines whether to build both the system and shadow registers (<i>bld_shad_reg</i> = 1) or just the system register (<i>bld_shad_reg</i> = 0)

a. The upper bound of the legal range is a guideline to ensure reasonable compile times.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
dw/sim_ver/DW04_shad_reg.v	Verilog simulation model source code

For normal operation, SE (scan enable) should be de-asserted (low). However, to scan out the contents of the shadow register, assert SE (high) and toggle shad_clk. The shadow register is then in shift register mode, with the most significant bit (MSB) driving SO, and data set-up on to the SI pin being clocked into the least significant bit (LSB).

Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)
```

The warning messages for this model include the following:

- If values other than 1 or 0 are present on a clock port, the following message is displayed:

```
WARNING: <instance_path>.<clock_name>_monitor:  
at time = <timestamp>, Detected unknown value, x, on <clock_name> input.
```

To suppress only this warning message for all DWBB components, use the following macro:

- Define the DW_DISABLE_CLK_MONITOR macro. You can define this macro in the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_DISABLE_CLK_MONITOR
```

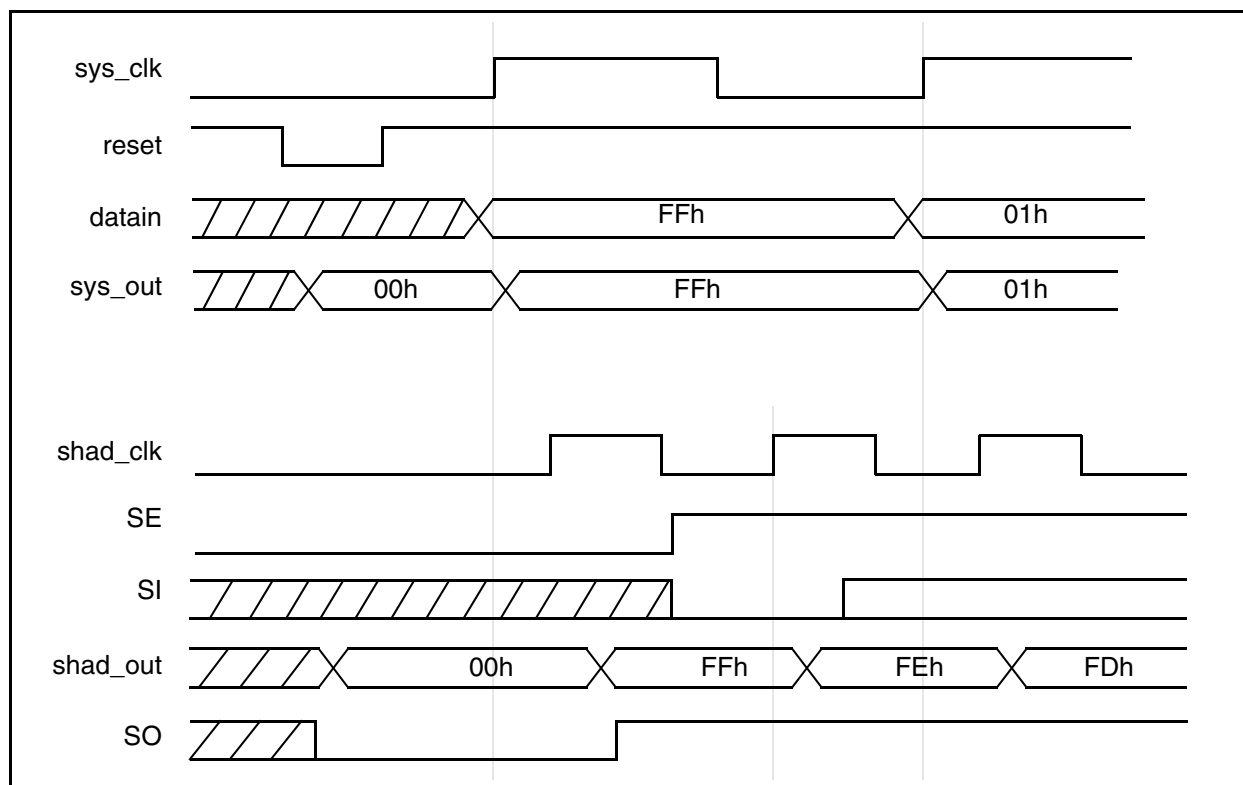
- Or, include a command line option to the simulator, such as:

```
+define+DW_DISABLE_CLK_MONITOR (which is used for the Synopsys VCS simulator)
```

This message is also suppressed using the DW_SUPPRESS_WARN macro explained earlier.

Timing Diagram

Figure 1-1 Timing Waveforms



Related Topics

- [Memory – Registers Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW04_shad_reg_inst is
  generic ( inst_width      : POSITIVE := 8;
            inst_bld_shad_reg : NATURAL := 1 );
  port ( inst_datain      : in std_logic_vector(inst_width-1 downto 0);
        inst_sys_clk     : in std_logic;
        inst_shad_clk    : in std_logic;
        inst_reset       : in std_logic;
        inst_SI          : in std_logic;
        inst_SE          : in std_logic;
        sys_out_inst     : out std_logic_vector(inst_width-1 downto 0);
        shad_out_inst    : out std_logic_vector(inst_width-1 downto 0);
        SO_inst         : out std_logic );
end DW04_shad_reg_inst;

architecture inst of DW04_shad_reg_inst is
begin

  --Instance of DW04_shad_reg
  U1 : DW04_shad_reg
    generic map ( width => inst_width,  bld_shad_reg => inst_bld_shad_reg )
    port map ( datain => inst_datain,  sys_clk => inst_sys_clk,
              shad_clk => inst_shad_clk,  reset => inst_reset,
              SI => inst_SI,  SE => inst_SE,  sys_out => sys_out_inst,
              shad_out => shad_out_inst,  SO => SO_inst );

end inst;

-- pragma translate_off
configuration DW04_shad_reg_inst_cfg_inst of DW04_shad_reg_inst is
  for inst
    end for; -- inst
end DW04_shad_reg_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW04_shad_reg_inst( inst_datain, inst_sys_clk, inst_shad_clk,
                          inst_reset, inst_SI, inst_SE, sys_out_inst,
                          shad_out_inst, SO_inst );

    parameter width = 8;
    parameter bld_shad_reg = 1;

    input [width-1 : 0] inst_datain;
    input inst_sys_clk;
    input inst_shad_clk;
    input inst_reset;
    input inst_SI;
    input inst_SE;
    output [width-1 : 0] sys_out_inst;
    output [width-1 : 0] shad_out_inst;
    output SO_inst;

    // Instance of DW04_shad_reg
    DW04_shad_reg #(width, bld_shad_reg)
        U1 ( .datain(inst_datain), .sys_clk(inst_sys_clk),
            .shad_clk(inst_shad_clk), .reset(inst_reset), .SI(inst_SI),
            .SE(inst_SE), .sys_out(sys_out_inst), .shad_out(shad_out_inst),
            .SO(SO_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
July 2020	DWBB_201912.5	■ Adjusted content and title of “ Suppressing Warning Messages During Verilog Simulation ” on page 3 and added the DW_SUPPRESS_WARN macro
October 2019	DWBB_201903.5	■ Added the “Disabling Clock Monitor Messages” section
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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